

The Hidden Costs of Urban Renewal

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Check the slides: mateomoglia.github.io/files/JMPMoglia_slides.pdf

Check the paper: mateomoglia.github.io/files/JMPMoglia.pdf

CREST PhD Seminar, mock job market talk — September 2026

Productive cities no longer grow outward

- ▶ Paris area, 1999–2021: population **+11%**, land take-up **+0.9%**
- ▶ Land at the fringe is exhausted. Growth happens **inside** the built envelope
- ▶ Same picture in Tokyo, New York, San Francisco, Seoul, Toronto (Frokling et al., 2025; Rollet, 2026)

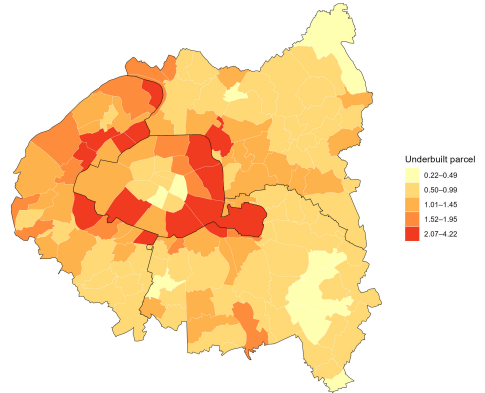
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The margin of adjustment is no longer vacant land. It is **redevelopment**.

Yet valuable land remains empty where rents are highest

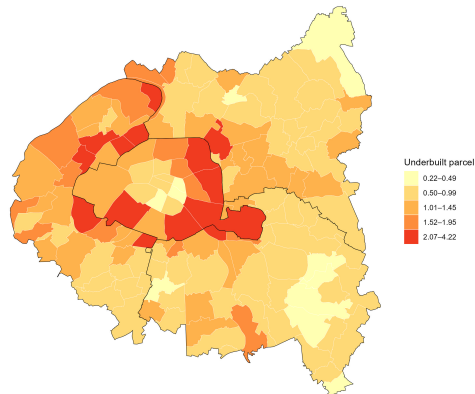
- ▶ 8,000+ parcels are empty, vacant, or *infills*
- ▶ All of them sit in the **98th percentile** of French rents
- ▶ Textbook urban economics says land moves to its highest-valued use



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High rents, high returns, **no redevelopment**



The standard answer is regulation: it is incomplete

Regulation as a supply tax (Glaeser & Gyourko, 2003, 2018; Gyourko & Molloy, 2015; Chapelle et al., 2025)

- Density caps, zoning, permitting, local opposition explain low supply elasticities in high-price cities

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- ▶ Density caps, zoning, permitting, local opposition explain low supply elasticities in high-price cities
- ▶ But a developer must acquire the land, bargain with the owner, evict, demolish, remediate, wait → high redevelopment fixed costs (Murphy, 2018; Peng, 2023; Rollet, JMP)
- ▶ These costs are **sunk**, **irreversible**, and **do not scale** with the building
 - Redevelopment is not only a production decision but also a timing decision
 - Fixed costs act as a barrier to entry

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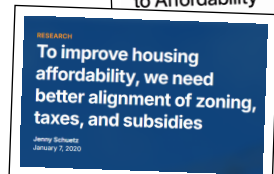
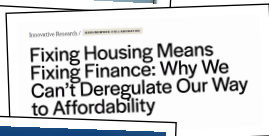
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A low redevelopment rate does not, by itself, imply stringent regulation

Why this matters now

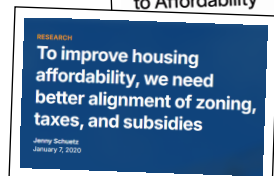
- ▶ **Zero net land take** (*Zéro Artificialisation Nette*) is a French and EU objective since 2021
 - Halve land take-up by 2031, reach net zero by 2050
 - Greenfield expansion is closed by law. Every additional unit must come from the existing stock
 - *Fonds Friches*: a nationwide brownfield cleaning programme, around **50 €/m²** of land
- ▶ Housing supply crisis is fuelling political discontent and calls for local leader action



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Should (local) governments **deregulate** or **subsidise** redevelopment?



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1. **How large** are the fixed costs of redevelopment?
2. **What are they made of?**
3. **How much of the low supply response** do they explain, relative to land-use regulation?

What I do

1. **Gather** a parcel-level panel of the Paris area, 2014–2025
 - 500,000+ developed parcels. Cadastre, permits, transactions, 3D envelopes, zoning, soil, and a developer cost survey
 - Unique granularity; **developer-reported costs**

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2. **Build** an option-value model of redevelopment with two frictions
 - Uncertainty comes from the **price process**, not from idiosyncratic profitability shocks
 - **Time-to-build** and **bargained land** taken to parcel-level data

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 - Net of the delay wedge and of the developer's rent, both calibrated outside the price equation
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 - Net of the delay wedge and of the developer's rent, both calibrated outside the price equation
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4. **Decompose** F_i against directly measured components, then run counterfactuals
 - Observable at the parcel level. Subsidies against density limit
 - Separate technological for regulation frictions; compare the margins

What I find

- ▶ **Fixed costs are large.** The median reachable parcel needs a transfer of 1,839 €/m² of land.
 - Same order as the cost of building the replacement, $c_0 = 1,186$ €/m².
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- ▶ **Regulation does not explain fixed costs alone.** Of the cross-sectional variance of F_i :
 - land-use regulation 21%, demolition 15%, surrounding 12%, and 37% unexplained
- ▶ **Fixed costs are the binding friction.**
 - Setting $F_i = 0$ raises supply by 104.1% and cuts rent by 22.7%.
 - Removing height limit closes only 41% of that gap

The binding constraint is not: **can I build?** but **how much does it cost** to build?

Related literature

1. **Structural housing supply and fixed costs**

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5. Land assembly and developer market power

Miceli & Sirmans (2007), Brooks & Lutz (2016), Quintero (2023), Charreire (2025), Lefebvre (2026), Favilukis, Pikulina, and Wu (2026)

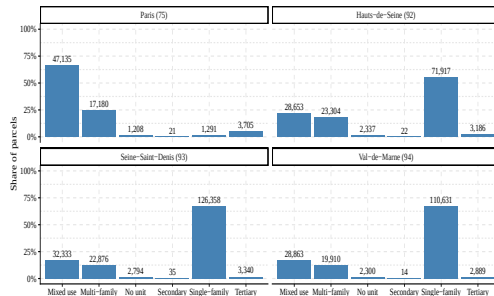
Roadmap

1. Introduction
2. Data and motivating facts
3. A dynamic model of redevelopment
4. Calibration
5. What is in the fixed cost?
6. Counterfactuals

Data and motivating facts

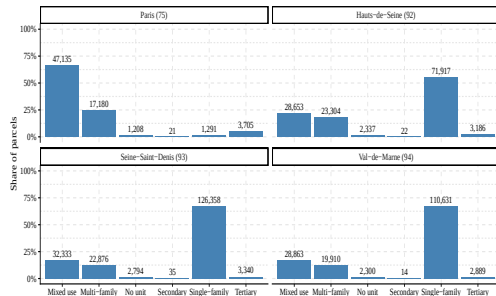
A (constructible) parcel-level dataset of the Paris area

- **Stock:** *Fichiers Fonciers* (fiscal universe of units), destination, floorspace, construction date (2014–2025)



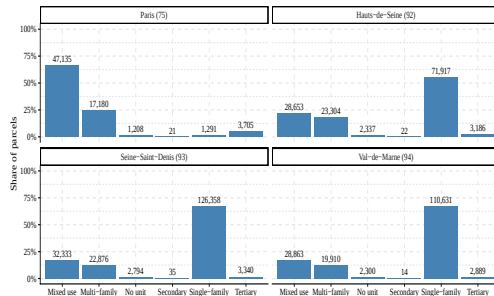
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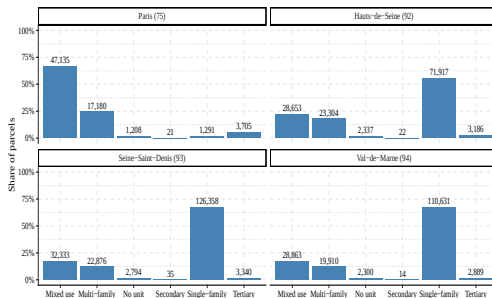
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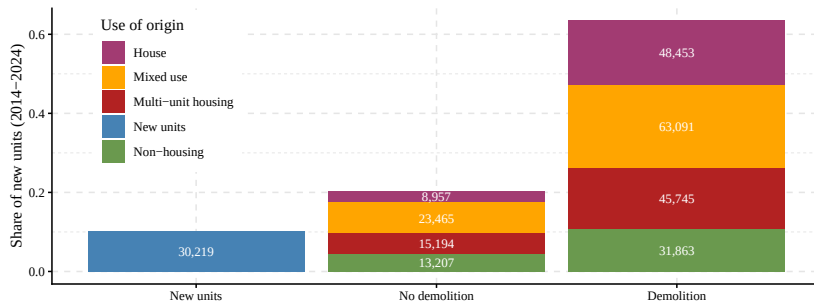
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Also: *BD Topo* 3D envelopes, *Apur* zoning, geographical and network data, *Fidéli* income.



Housing is supplied by replacing what already stands



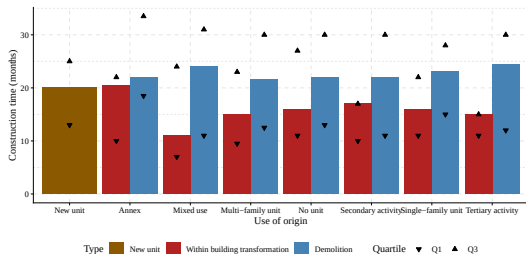
- ▶ 18,873 permits, 298,329 new units, 18.0 million m² (2014-2024).
- ▶ **64%** of new units come from demolition-reconstruction, 20% from transformation within the envelope, 10% from empty parcels.
- ▶ A third of new units are supplied on **former non-housing** parcels

Two model frictions are visible in the data

Concentration

Assembly premium

Construction takes time



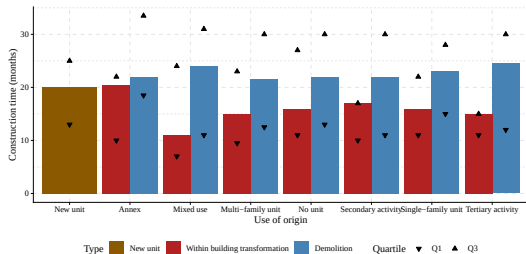
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- Demolition is slower than within-building transformation

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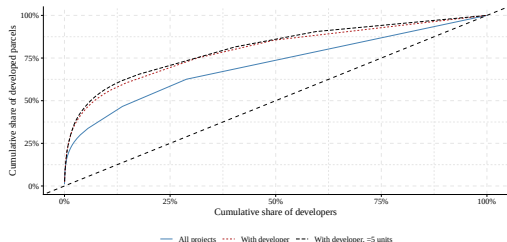
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Land is bought in a thin local market



- Median municipal market: an effective **17** developers (low competition on land)
- Largest project: 1.35% of the municipal stock (atomistic on prices)

Land-use regulations are three-fold

D: in the data; **M**: in the model; **U**: unobserved

1. Nationwide regulations

- D Historical protections (500m around a building)
- U Environmental concerns (specific materials)
- U Accessibility mandates, parking, etc.

2. Local regulations

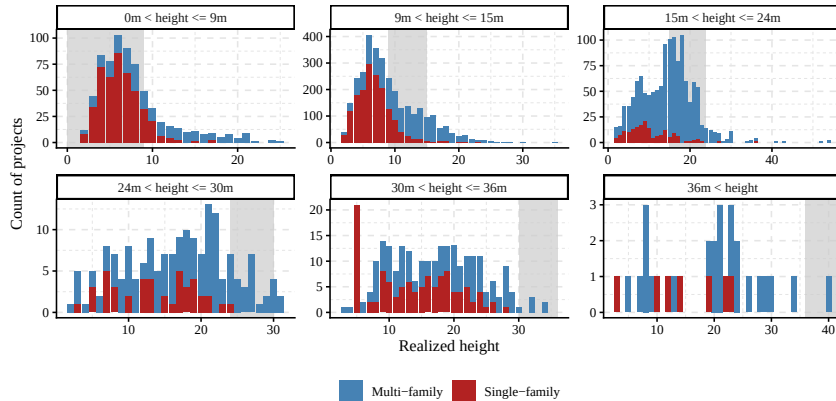
- M Maximum height
- D Minimum lot size, floor-to-area ratio
- U Building composition
- D Zoning
- D Discretionary mayor power

3. Local political economy

- D Building permit appeals
- U Citizen push-backs, NIMBY

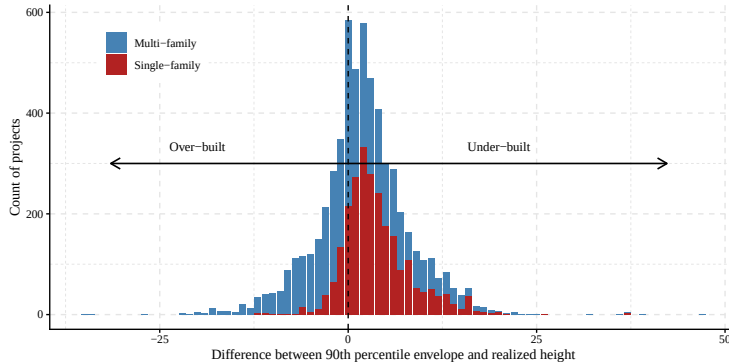
New constructions do not all reach the regulatory cap

Comparison with regulatory maximum height



New constructions do not all reach the regulatory cap

Comparison with 90th percentile building envelope around



A dynamic model of redevelopment

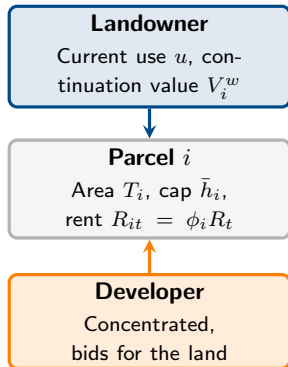
Main objectives

1. Recover **a redevelopment threshold** R_i^* , the rent above which parcel i is redeveloped
 - The incumbent landowner earns a rental income that depends on the current use (housing, non-housing, vacant)
 - She sells once the price offered exceeds the value of holding on
 - That indifference point defines R_i^*

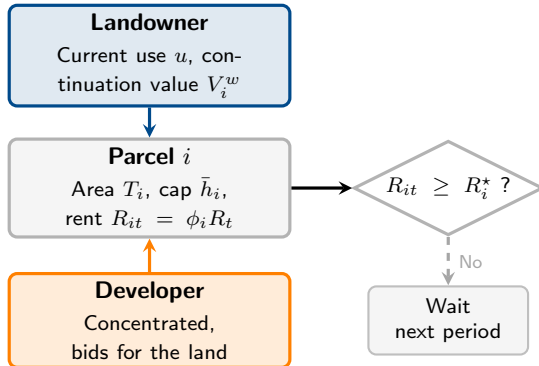
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2. Extract the **redevelopment fixed cost** F_i
 - Conditional on buying, the developer chooses $h^* \leq \bar{h}_i$ to maximise profit
 - F_i is what separates the profit she can make from the price she pays

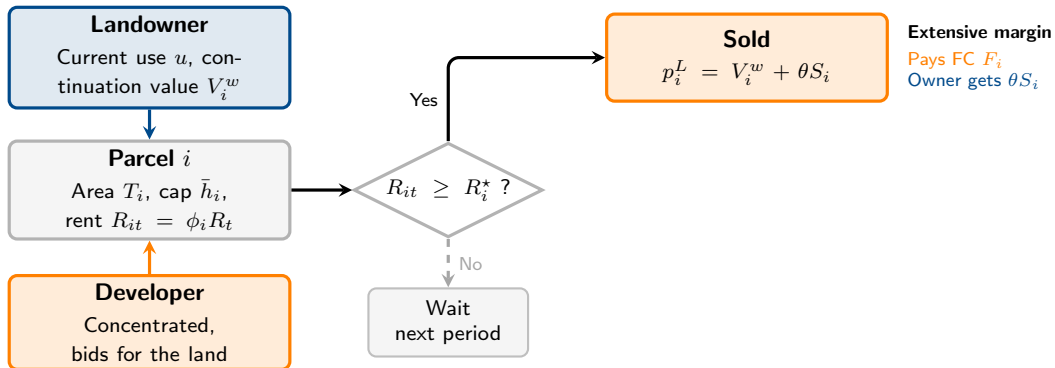
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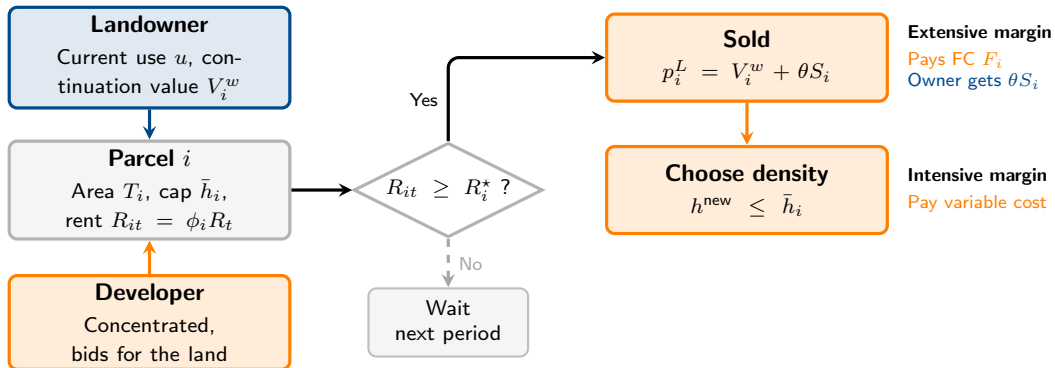
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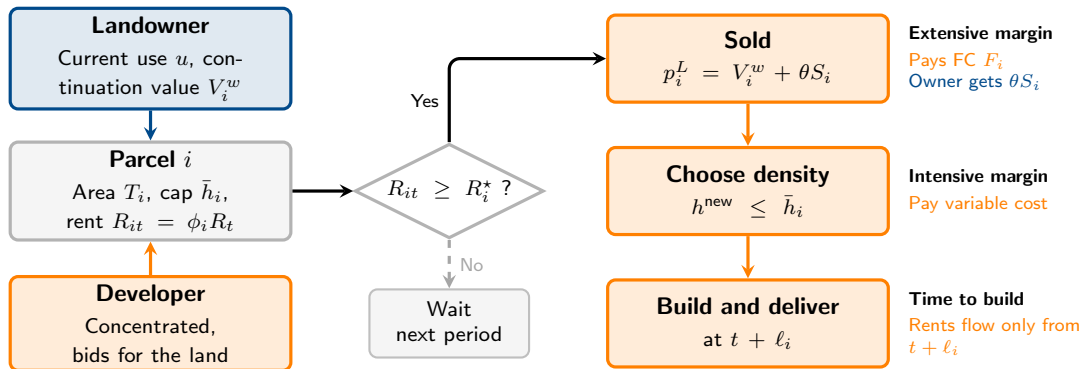
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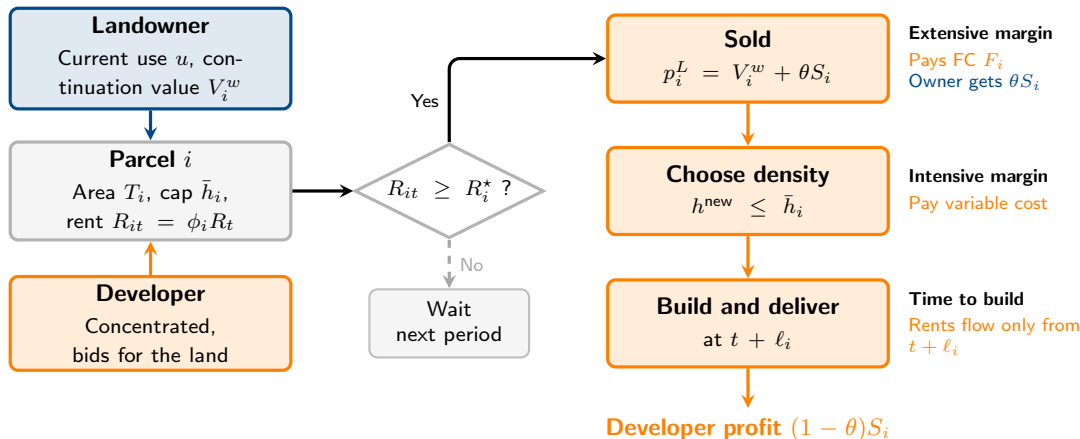
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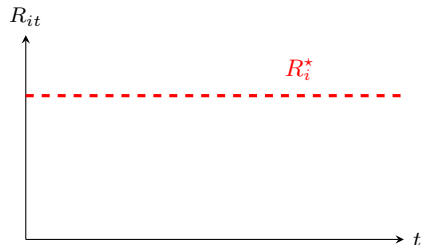


Surplus $S_i(R_t) = \pi_i(R_{it}) - F_i - V_i^w$, split θ . Trade occurs whenever $S_i \geq 0$.

Why does anyone wait?

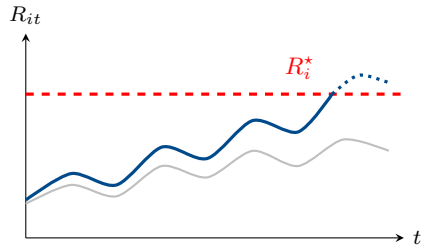
► Waiting is valuable

- **Irreversible**: demolition cannot be undone, F_i is sunk
- **Uncertain**: nobody knows the rent in ten years
- **No deadline**: the parcel does not expire



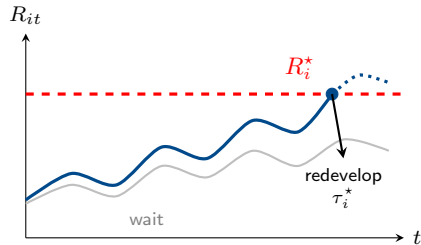
Why does anyone wait?

- ▶ Waiting is valuable
- ▶ **Rents follow a geometric Brownian motion**
 - Drift up at δ per year
 - Hit by proportional shocks of scale σ



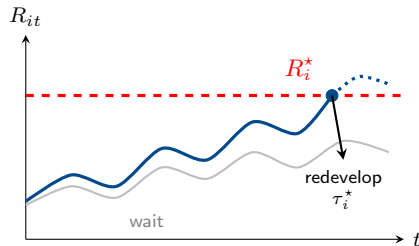
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- ▶ Waiting is valuable
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- ▶ **Building today locks** in today's rent forever
- ▶ **Waiting keeps** the upside and **discards** the downside
 - **Landowner**: keeps V_i^w and the right to sell later
 - **Developer**: avoids sinking F_i into a project a lower rent would not support



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 - **Developer**: avoids sinking F_i into a project a lower rent would not support
- ▶ They agree on the date. Both take a fixed share of the same surplus, so **both want it maximised**



The trigger is a **markup** over break-even: $R_i^* = \frac{\beta}{\beta-1} R_i^{\text{naive}}$, with $\frac{\beta}{\beta-1} = 1.50$

Intensive margin: how much to build?

A developer on parcel i chooses density to maximise value created, net of convex construction cost:

$$\pi_i(R_{it}) = \max_{h^{\text{new}} \leq \bar{h}_i} \underbrace{\lambda_i^R \frac{R_{it} h^{\text{new}}}{\rho - \delta}}_{\text{Delayed revenue}} - \lambda_i^C \text{ Convex variable cost}$$

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- ▶ Two wedges summarise the lag: $\lambda_i^R = e^{-(\rho-\delta)\ell_i}$ and $\lambda_i^C = (1 - e^{-\rho\ell_i})/(\rho\ell_i)$. Only their ratio κ_i matters
- ▶ Optimal density is linear in parcel area and increasing in rent:

$$h_i^{\text{new}} = \left[\frac{\zeta \kappa_i R_{it}}{c_0(\zeta + 1)(\rho - \delta)} \right]^\zeta T_i \quad \text{if } R_{it} < \bar{R}_i, \quad \bar{h}_i \text{ otherwise}$$

- ▶ Delay makes **projects smaller** (by κ_i^ζ) and **widens** the interior regime (saturation rent rises by κ_i^{-1}).

Extensive margin I: land is bargained, not auctioned

- The landowner's outside option is the value of the **current use** V_i^w

$$\text{Housing: } V_i^w = \frac{R_{it}h_i^{\text{old}}}{\rho - \delta}, \quad \text{Non-housing: } \frac{v_i^C}{\rho - \delta^C}, \quad \text{Vacant: } 0$$

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- ▶ The **surplus** created by the transaction is $S_i(R_t) = \pi_i(R_{it}) - F_i - V_i^w$
- ▶ Generalised Nash with landowner weight θ gives the price and the developer's rent

$$p_i^L = V_i^w + \theta S_i, \quad \Pi_i^D = (1 - \theta)S_i$$

Extensive margin I: land is bargained, not auctioned

- ▶ The landowner's outside option is the value of the **current use** V_i^w
- ▶ The **surplus** created by the transaction is $S_i(R_t) = \pi_i(R_{it}) - F_i - V_i^w$
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- ▶ $\theta = 1$ recovers the **competitive benchmark**: zero developer profit, full capitalisation into land
- ▶ $\partial p_i^L / \partial F_i = -\theta$: landowners bear only a **fraction** of transition costs
- ▶ Density is unaffected. Both parties take a constant share of the maximised surplus

Extensive margin II: the stopping rule

- The landowner solves an optimal stopping problem, $\tau_i^* = \inf\{t : R_t \geq R_i^*\}$, with

$$R_i^* = \underbrace{\frac{\beta}{\beta-1}}_{\text{Option multiplier}} \cdot \underbrace{\frac{1}{\kappa_i} \frac{(\zeta+1)(\rho-\delta)}{\zeta} \left[\frac{\zeta(F_i + V_i^w)}{\lambda_i^C c_0^{-\zeta} T_i} \right]^{\frac{1}{\zeta+1}}}_{\text{Naive break-even rent}}$$

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- $\beta > 1$ solves $\frac{1}{2}\sigma^2\beta(\beta - 1) + \delta\beta - \rho = 0$: it depends on σ , δ and ρ **only**
 - More volatility σ or a higher discount rate ρ raises the trigger. Faster rent growth δ lowers it
 - The parcel enters through F_i , V_i^w , T_i and $\bar{h}_i$, never through the multiplier

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Proposition 2. Bargaining is neutral for timing. The developer maximises $(1-\theta)S_i$ and $(1-\theta)$ is a constant, so R_i^* is invariant to θ .

A fixed cost that is really a fixed cost

A structural residual is only as interpretable as the things it has been **purged of**

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 - the **developer's rent**: land does not trade at its full redevelopment value

A fixed cost that is really a fixed cost

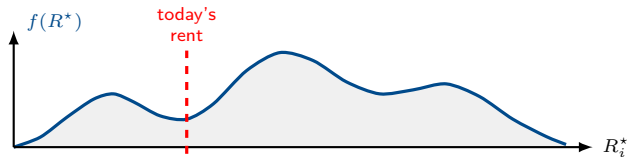
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 - the **delay wedge**: a project costs now and earns only from $t + \ell_u$
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- ▶ Option value is **not** netted out, and should not be
 - It is not a cost. It is what makes waiting optimal *at a given cost*

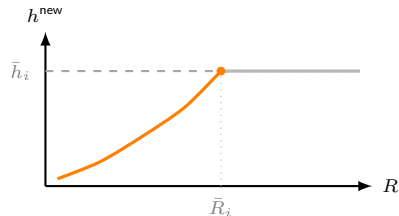
What remains is euros that must be spent before the first square metre is built

What governs the supply elasticity

$$\frac{dQ_t}{dR_t} = \underbrace{E\left[\frac{\partial h_i^{\text{new}}}{\partial R} \mid R_i^* = R_t\right]}_{\text{How much \textcolor{blue}{more} each builds}} \cdot f(R_t) + \underbrace{E[h_i^{\text{new}}(R_t) \mid R_i^* = R_t]}_{\text{How much \textcolor{blue}{each} builds}} \cdot \underbrace{f(R_t)}_{\text{How \textcolor{blue}{many} cross}}$$



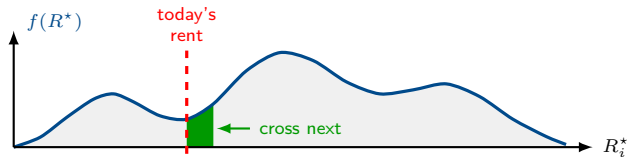
Extensive: how much mass sits at today's rent



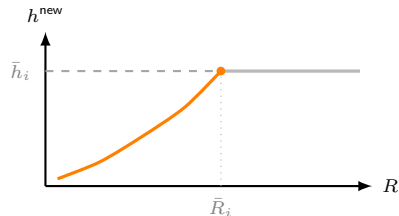
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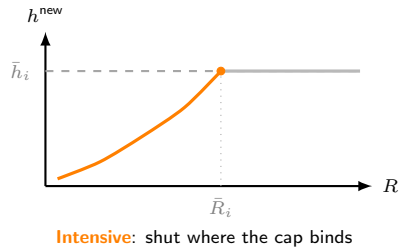
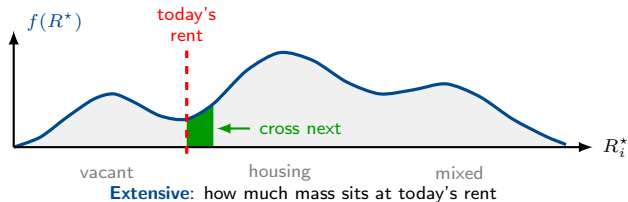
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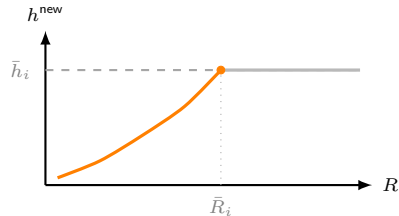
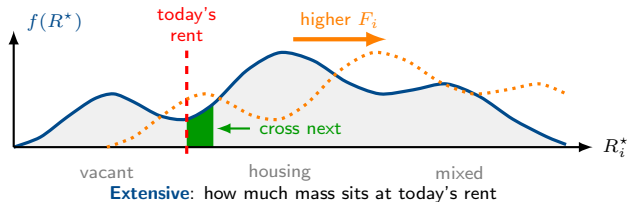
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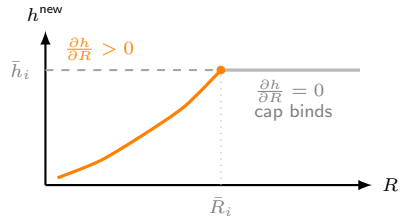
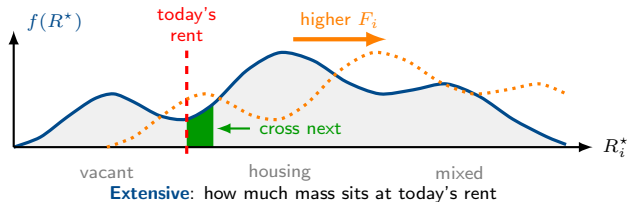


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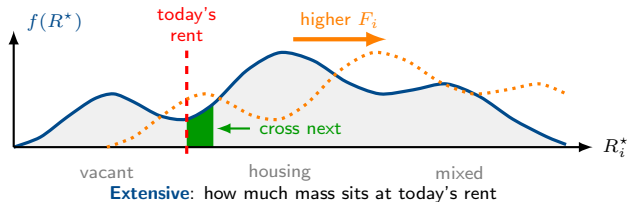


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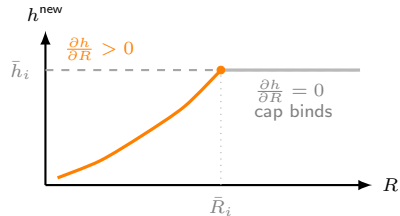
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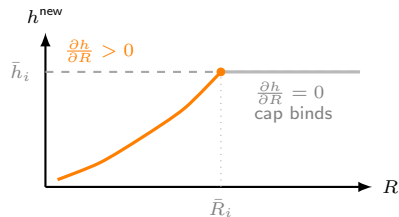
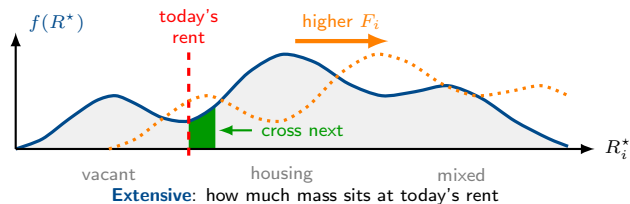


Intensive: shut where the cap binds

- **The cap** shuts the orange term
- **The lag** raises triggers but pulls parcels out of the corner

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Intensive: shut where the cap binds

Fixed costs and regulation act on **different margins**

Calibration

Every parameter is fixed outside the price equation

Par.	Name	Source	Value
μ	Demand growth	Obs. Loyers, FF	0.01
σ	Demand volatility	Obs. Loyers, FF	0.04
α	Inverse demand elast.	Chapelle et al. (2023)	0.35
δ	Perceived rent growth	ρ — net yield (IDF)	0.02
ρ	Discount rate	Banque de France	0.065
β	Option root	Computed	2.98
ζ	Cost convexity	Combes et al. (2021)	1.17
c_0	Construction cost	PRLN, EPTB	1,186
ℓ_{hous}	Lag, housing	Sitadel	2.0
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θ	Landowner weight	$1 - \text{HHI}$, floor $F_i \geq 0$	1.00
F_i	Fixed cost	Structural residual	—
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► The option multiplier is
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- The option multiplier is $\beta/(\beta - 1) = 1.50$
- **Demolition is measured**, not assumed
 - 35–56 €/m² by destination, plus a 25,269 € mobilisation cost.
 - Uncorrelated with authorised floorspace (fixed in exactly the model's sense)

Demolition costs

Four over-identification checks

1. **Supply elasticity**. The model delivers 0.6 against an external estimate of 0.35 (Chapelle et al., 2023)

θ floor

(ℓ, θ) grid

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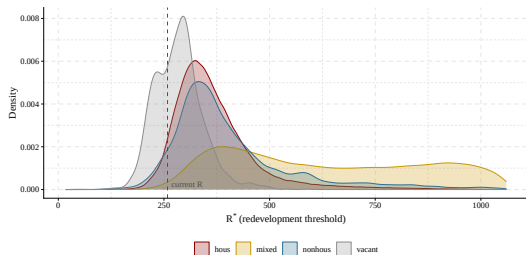
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4. **The sign restriction disciplines θ** . The share of negative F_i is monotone in θ , which delivers a floor of 0.38

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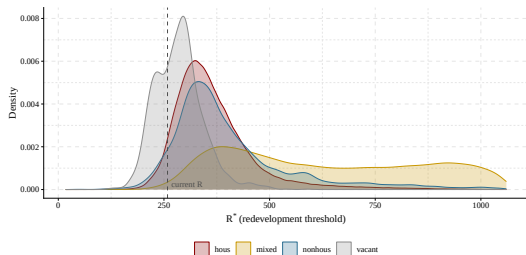
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Taking stock: where the parcels sit

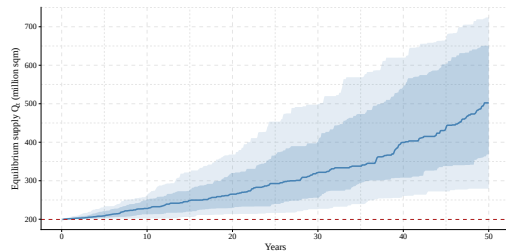


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- ▶ Supply grows slowly along the demand gradient over 50 years.
- ▶ At the 2023 rent, the elasticity of 0.63 is **almost entirely extensive** in Paris. The cap binds for most parcels.

Maps

Regime split

What is in the fixed cost?

First, what F_i does not contain

Use	N	Median, EUR per sqm of parcel			Share of aggregate gap		
		Delay	Developer rent	Fixed cost	Delay	Developer rent	Fixed cost
Housing	104,760	608	0	3,282	16.0%	0.0%	84.0%
Mixed	16,339	1,020	0	4,457	17.5%	0.0%	82.5%
Non-housing	4,950	891	0	4,279	17.4%	0.0%	82.6%
Vacant	1,577	529	0	3,372	13.4%	0.0%	86.6%

- ▶ Across the sample the construction lag takes 16.8% of the gap and the developer's rent 0.0%
- ▶ The fixed cost is the remaining 83.2%, and that is the object I decompose
- ▶ The delay wedge is largest for non-housing origins: longest lags, largest redevelopment values

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The magnitudes are shares of a **smaller total** than a competitive, instantaneous-construction benchmark would give

Decomposing the fixed cost

$$\log \frac{F_i}{T_i} = a + \underbrace{\mathbf{Dem}_i}_{\text{Demolition}} + \underbrace{\mathbf{Soil}_i + \mathbf{Conn}_i + \mathbf{Surr}_i + b \log T_i}_{\text{Parcel situation}} + \underbrace{\mathbf{Reg}_i}_{\text{Regulation}} + \mathbf{Mun}_i + e_i$$

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- ▶ Cluster bootstrap on municipalities, 500 replications

Demolition costs

Coefficients

Pre-conditions

Regulation makes fixed costs *differ*. It does not make them *large*

Component	Share of total variance (%)		
	With controls	Without controls	Within commune
Demolition	11.4 (1.0)	14.8 (1.3)	2.0 (0.2)
Ground condition	9.2 (1.5)	14.1 (2.1)	3.3 (1.0)
Grid connectivity	0.3 (0.6)	0.4 (0.8)	0.2 (0.3)
Built surroundings	9.7 (1.1)	12.5 (1.3)	1.0 (0.1)
Parcel size	0.4 (0.1)	0.6 (0.1)	0.0 (0.0)
Land-use regulation	14.9 (1.1)	20.6 (1.6)	9.8 (0.9)
Municipal (location)	24.4 (1.4)	–	–
Unexplained	29.7 (3.5)	37.0 (3.8)	83.6 (1.2)

Robustness

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Unexplained	29.7 (3.5)	37.0 (3.8)	83.6 (1.2)

► **Regulation** leads ahead of demolition alone

- Ranking holds across the whole parameter band and every price specification

Robustness

Regulation makes fixed costs *differ*. It does not make them *large*

Component	Share of total variance (%)		
	With controls	Without controls	Within commune
Demolition	11.4 (1.0)	14.8 (1.3)	2.0 (0.2)
Ground condition	9.2 (1.5)	14.1 (2.1)	3.3 (1.0)
Grid connectivity	0.3 (0.6)	0.4 (0.8)	0.2 (0.3)
Built surroundings	9.7 (1.1)	12.5 (1.3)	1.0 (0.1)
Parcel size	0.4 (0.1)	0.6 (0.1)	0.0 (0.0)
Land-use regulation	14.9 (1.1)	20.6 (1.6)	9.8 (0.9)
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 - Ranking holds across the whole parameter band and every price specification
- ▶ **Parcel situation** is second and connectivity indistinguishable from zero
- ▶ **37.0%** is explained by nothing observable
 - Local political economy

Robustness

Counterfactuals

Two families of instruments, two margins

Acting on F_i

Acting on $\bar{h}_i$

Two families of instruments, two margins

Acting on F_i

- ▶ $F_i = 0$: a **bound**, not a policy
- ▶ Subsidise **under-built** parcels within 1 km of Paris (38k parcels, 7% of stock)
- ▶ Subsidise **vacant and non-housing** parcels (10k parcels, 1.8%)
 - $s \in \{50, 100, 500, 1,000\} \text{ €/m}^2$, capped at F_i . The floor is the *Fonds Friches* rate.

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Acting on $\bar{h}_i$

- ▶ **Remove the cap** entirely ($\bar{h}_i \rightarrow \infty$): a bound on any relaxation
- ▶ **Raise every** below-median municipality to the **metropolitan median FAR** of 1.9
 - The only implementable one of the three

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All equilibria re-solved under the **same draws** of A_t . I report the partial-equilibrium shift and the general-equilibrium response separately.

Fixed costs are the binding friction

Family	Scenario	ΔQ (GE)	ΔQ (PE)	ΔR	Projects	Net floor	Outlay
Fiscal	No fixed cost	104.1%	148.1%	-22.7%	193,331	282.59	–
	Underused, near Paris	2.3%	5.2%	-0.8%	-2,215	6.35	6,515
	Non-housing and vacant	2.3%	5.0%	-0.7%	-3,327	6.13	10,719
Regulatory	Density cap lifted	21.6%	61.3%	-6.5%	-35,478	58.74	–
	Below-median to median	9.9%	21.1%	-3.1%	-19,638	26.79	–

PE/GE path

Stock composition

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 - Equivalent to a uniform subsidy of 1,343 €/m²
- ▶ The cap binds for 76.6% of parcels, so raising it leaves the rest untouched

Fixed costs are the binding friction

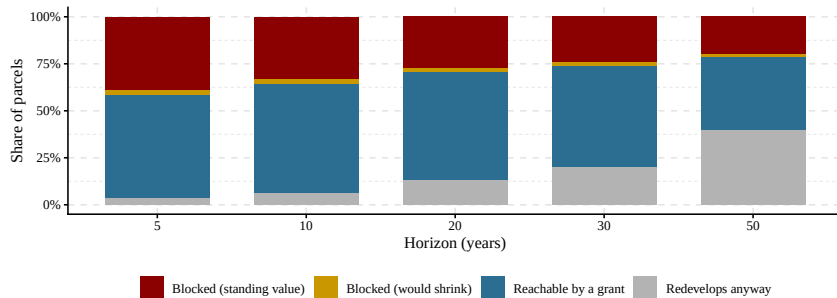
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282m square meters (/ 60 sqm) : 67 years of Région Île de France target of 70,000 new units per year

PE/GE path

Stock composition

How large a transfer would tip the median parcel?

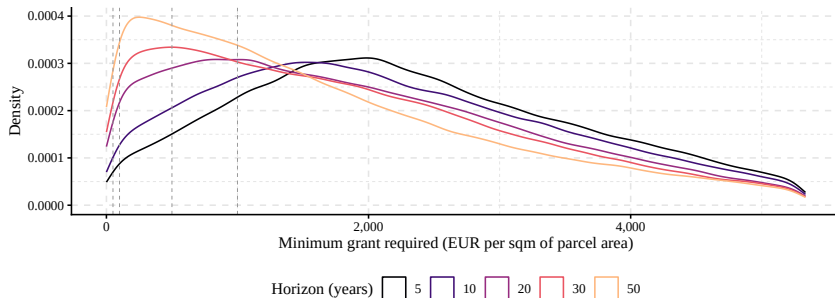


- At twenty years: 13.4% redevelop anyway, 57.1% are **reachable**, 27.3% blocked by standing value, 2.2% by the cap.

Take-up vs conversion

Budget frontier

How large a transfer would tip the median parcel?



- Median inducing subsidy: **1,839 €/m²** of land
 - Against $c_0 = 1,186$ €/m² of floorspace
- A 50 €/m² transfer reaches 1.2% of the reachable set

Targeting on observables is expensive

- ▶ At 50 €/m² the under-built scheme converts 0.38% of targeted parcels; the non-residential scheme 0.67%.
 - Twentyfold escalation to 1,000 €/m² reaches 8.6% and 19.6%.

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 - Twentyfold escalation to 1,000 €/m² reaches 8.6% and 19.6%.
- ▶ **The inframarginal share is invariant to the subsidy.** Lowering R_i^* cannot stop a parcel from crossing.
 - At 50 €/m², 95.7% of the outlay goes to parcels that would have redeveloped anyway.

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- ▶ **The inframarginal share is invariant to the subsidy**. Lowering R_i^* cannot stop a parcel from crossing.
 - At 50 €/m², 95.7% of the outlay goes to parcels that would have redeveloped anyway.
- ▶ **Targetting** the parcels **most likely to redevelop** is not efficient
 - It maximises the transfer and minimises the causal effect
 - The model helps screening the relevant parcels

Conclusion

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 - Removing F_i : +104.1% supply. Removing the cap entirely: 41% of that

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Conclusion

- ▶ Where cities grow inward, the binding constraint is the **cost of clearing the site**, not the permitted envelope
 - Removing F_i : +104.1% supply. Removing the cap entirely: 41% of that
- ▶ Regulation still matters explains only 21% of variance
- ▶ Calls for policy actions on fixed costs!
- ▶ **Next steps:**
 - **Fiscal incidence**: optimal subsidy under budget constraint and tax revenues
 - Counterfactuals on the **option value** itself: preferred-rate credit, uncertainty reduction
 - Rent control and social housing mandates
 - Estimate ζ directly on multi-family projects

The Hidden Costs of Urban Renewal

Matéo Moglia

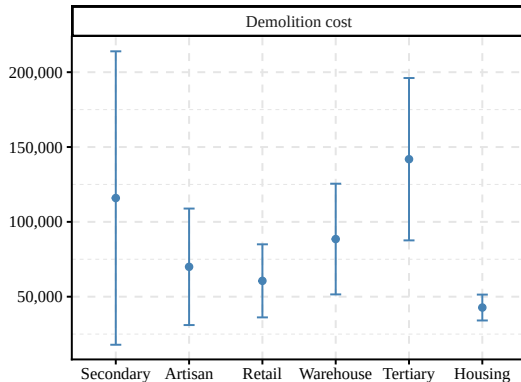
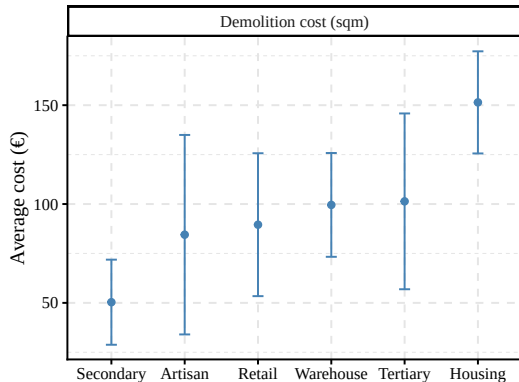
CREST - IPParis

Check the slides: mateomoglia.github.io/files/JMPMoglia_slides.pdf

Check the paper: mateomoglia.github.io/files/JMPMoglia.pdf

Appendix

Demolition costs, measured

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Demolition costs by project size

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	Demolition cost (1)
Authorised floorspace	31.3* (16.3)
Number of dwellings	-1,081.5 (1,005.1)
Observations	341
R ²	0.330
Year fixed effects	✓
Region fixed effects	✓

Developer concentration in land acquisition

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Sample

Statistic	All	With dev.	Dev. & large (≥ 5)
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Municipal

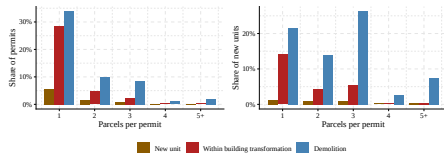
Active developers	101	35	25
HHI	2.00	5.79	7.44
CR4	18.6%	35.9%	42.3%
Effective devs.	50.0	17.3	13.4

Department

Active developers	4,784	1,360	916
HHI	0.15	0.60	0.85
CR4	5.5%	11.4%	13.8%
Effective devs.	661.2	171.5	120.5

Regional

Active developers	16,810	4,249	2,789
HHI	0.07	0.30	0.43
CR4	3.4%	7.2%	8.7%
Effective devs.	1452.2	332.8	233.4



- Concentration is local and dissolves with aggregation: CR4 falls from 35.9% to 7.2%.
- Atomistic on the stock, not on a given parcel.

Share of aggregate stock

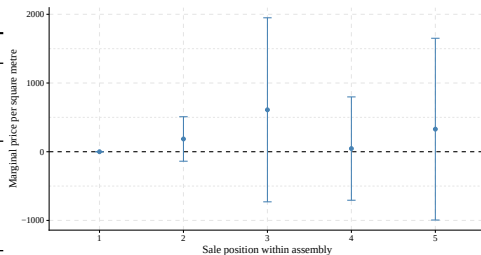
Share of aggregate flow

Level	Median	P99	Max	Median	P99	Max
Municipal	1.35%	6.19%	11.83%	14.60%	74.91%	76.91%
Department	0.10%	0.12%	0.12%	0.98%	2.09%	2.12%
Regional	0.02%	0.02%	0.02%	0.29%	0.29%	0.29%

Land assembly: a bundling premium, no hold-out escalation

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	(1)	(2)	(3)	(4)	(5)
Naive assembly	235.787** (98.821)	234.827** (98.988)	219.760** (90.048)	229.435** (88.396)	204.484** (87.811)
Sequential assembly	-3.618 (114.670)	2.601 (114.537)	83.677 (98.702)	102.256 (96.897)	73.244 (101.757)
Year FE	✓	✓	✓	✓	✓
Municipality FE	✓	✓	✓	✓	✓
Built area		✓	✓	✓	✓
Seller			✓	✓	✓
Buyer				✓	✓
Parcel count					✓
Mean Dep. Var	1,984	1,984	1,984	1,984	1,984
Observations	9,482	9,482	9,482	9,482	9,482
R ²	0.533	0.534	0.547	0.554	0.554



Time-to-build wedges

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Use of origin	ℓ_u	λ_u^C	λ_u^R	κ_u	Effect on		
					$\Delta\pi$	Δh^*	ΔR^*
Housing	2.00	0.9377	0.9139	0.9746	-11.3%	-3.0%	5.7%
Mixed	2.25	0.9303	0.9037	0.9714	-12.6%	-3.3%	6.4%
Non-housing	2.25	0.9303	0.9037	0.9714	-12.6%	-3.3%	6.4%
Vacant	1.75	0.9452	0.9243	0.9778	-10.0%	-2.6%	5.0%

- ▶ κ_u is far closer to one than λ_u^R : spreading the spend offsets most of the delay.
- ▶ Housing parcels: payoff -11.3% , density -3.0% , trigger $+5.7\%$.
- ▶ Second-order at parcel level. First-order in the aggregate, through the delivery lag.

The bargaining weight and its floor

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Use of origin	N	θ_i		
		Median	p75	p95
Housing	344,803	-0.261	0.068	0.370
Mixed	47,898	-0.319	0.135	0.389
Non-housing	6,873	0.201	0.321	0.508
Vacant	1,600	0.372	0.464	0.624

- ▶ $\theta_i = (p_i^L - V_i^w)/(\pi_i - V_i^w)$ is computable without knowing F_i .
- ▶ Any θ below the 95th percentile implies more than 5% of parcels with a negative fixed cost.
- ▶ The floor is $\theta \geq 0.38$. Adopted value: 1.00.

Robustness to the two frictions

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Regime split: where the density cap binds

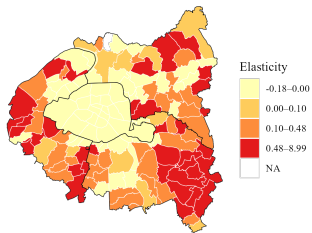
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Use	Dept.	Share of parcels		
		N	Interior share	$\Pr(\pi < 0)$
Housing	75	10,140	98.9%	0.19%
	92	95,953	26.6%	0.02%
	93	157,056	17.4%	0.01%
	94	136,295	16.2%	0.02%
Mixed	75	55,189	99.6%	0.05%
	92	28,575	66.8%	0.12%
	93	26,182	54.0%	0.32%
	94	24,582	48.4%	0.07%
Non-housing	75	1,472	86.0%	9.71%
	92	2,209	63.2%	3.67%
	93	2,613	48.2%	2.95%
	94	2,094	41.5%	1.77%
Vacant	75	42	85.7%	11.90%
	92	541	28.8%	1.11%
	93	672	22.8%	0.15%
	94	359	21.2%	0.56%

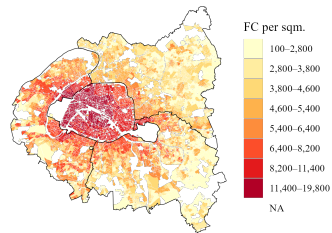
Elasticities and fixed costs in space

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Municipal supply elasticity



Fixed cost per m², cadastral section



Fixed cost decomposition: coefficients

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	log(Fixed cost per sqm)				Predicted cost	+ FE
	(1)	(2)	(3)	(4)	(5)	(6)
notationCE	0.046 (0.080)	0.045 (0.077)	0.019 (0.068)	0.051 (0.078)	0.020 (0.068)	-0.117** (0.056)
notatione2Cr-BE	0.271*** (0.080)	0.251*** (0.076)	0.224*** (0.080)	0.265*** (0.080)	0.229*** (0.081)	0.171 (0.146)
notatione4AM	0.282*** (0.100)	0.278*** (0.097)	0.244*** (0.084)	0.272*** (0.101)	0.238*** (0.085)	0.108 (0.104)
notatione4AP	0.058 (0.114)	0.060 (0.106)	0.047 (0.093)	0.064 (0.110)	0.050 (0.095)	-0.083 (0.060)
notatione4GS	0.322*** (0.082)	0.304*** (0.079)	0.249*** (0.073)	0.313*** (0.081)	0.250*** (0.073)	0.004 (0.063)
notatione4SC-AH	0.304*** (0.084)	0.286*** (0.092)	0.232*** (0.099)	0.289*** (0.082)	0.231** (0.092)	-0.008 (0.058)
notatione5CG	0.086 (0.077)	0.090 (0.075)	0.078 (0.062)	0.090 (0.075)	0.077 (0.061)	-0.017 (0.052)
notatione5MC	0.104 (0.087)	0.095 (0.083)	0.071 (0.071)	0.107 (0.085)	0.076 (0.071)	-0.068 (0.057)
notatione6-7MGC	0.017 (0.076)	0.016 (0.074)	0.024 (0.067)	0.022 (0.075)	0.027 (0.066)	-0.084 (0.056)
notatione6CSO	0.004 (0.079)	0.010 (0.077)	0.011 (0.067)	0.011 (0.077)	0.012 (0.066)	-0.089 (0.058)
notatione6SB-A	0.064 (0.078)	0.057 (0.075)	0.054 (0.064)	0.066 (0.077)	0.059 (0.064)	-0.069 (0.054)
notatione6SM	0.129 (0.084)	0.121 (0.081)	0.105 (0.070)	0.131 (0.083)	0.109 (0.070)	-0.027 (0.063)
notatione7C	0.118 (0.274)	0.106 (0.272)	0.056 (0.284)	0.134 (0.274)	0.068 (0.286)	-0.193 (0.186)
notatione7CCCh-MP	-0.082 (0.110)	-0.078 (0.109)	-0.073 (0.109)	-0.073 (0.110)	-0.070 (0.109)	-0.082 (0.064)
notatione7G	-0.001 (0.000)	-0.001 (0.000)	-0.002 (0.000)	0.005 (0.000)	0.001 (0.000)	-0.122** (0.000)

Pre-conditions: orthogonality and separability

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- ▶ **Orthogonality.** Regressing F_i/T_i on parcel area and the density cap gives an adjusted R^2 of 2%.
 - Correlation with area is essentially zero; with the cap, 0.098.
- ▶ **Separability.** The demolition-by-regulation interaction is rejected ($p = 0.003$), as is the omnibus test ($p = 0.001$).
 - It fails by a negligible margin: interactions raise R^2 by 0.0046 on half a million observations.
- ▶ **Normalisation.** F_i is taken per m^2 so the decomposition recovers economic content, not parcel size. Scale ends up at 0.6%.

Shapley shares: parameter band and price specification

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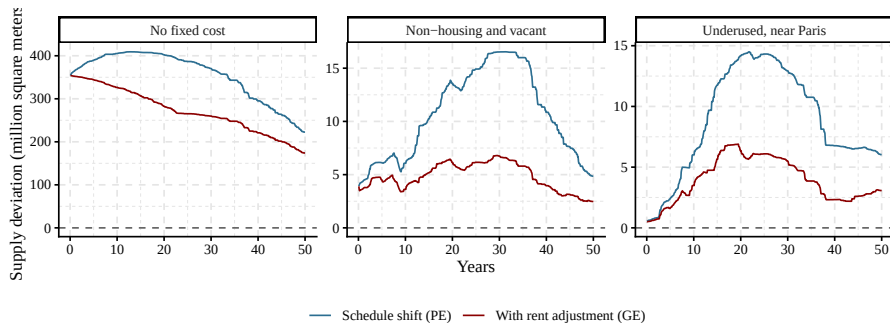
Component	Share of total variance (%), [min, max]	
	With controls	Without controls
Demolition	[8.8, 10.4]	[11.3, 13.8]
Parcel characteristics	[14.2, 17.6]	[19.7, 26.2]
Land-use regulation	[11.8, 17.2]	[17.5, 24.3]
Municipal (location)	[19.4, 26.8]	–
Unexplained	[28.0, 45.9]	[35.7, 51.5]

Component	Share of total variance (%), [min, max]	
	With controls	Without controls
Demolition	[10.3, 13.6]	[13.7, 17.6]
Parcel characteristics	[16.8, 17.5]	[24.6, 25.8]
Land-use regulation	[14.5, 16.3]	[20.7, 23.3]
Municipal (location)	[24.4, 26.2]	–
Unexplained	[29.5, 30.7]	[37.0, 37.5]

- ▶ Regulation 18–24 %, demolition 11–14 %. The intervals never meet.
- ▶ Unexplained share moves 36 –51 % across parameters but only 37–37 % across price specifications.

Partial and general equilibrium paths

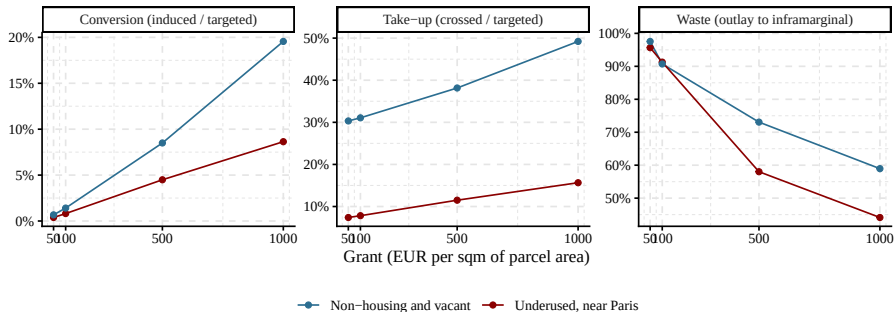
◀ Back



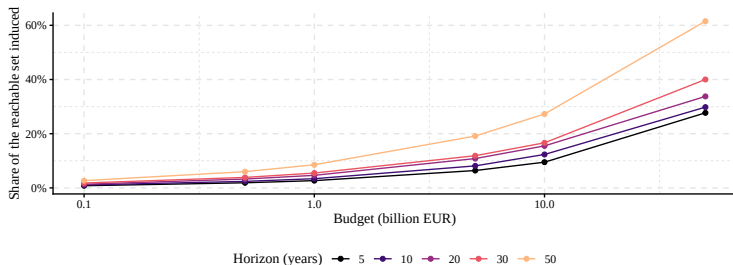
- ▶ The schedule shift peaks early and declines: the baseline rent eventually does the work by itself
- ▶ Argues for a **front-loaded** rather than a permanent instrument

Take-up, conversion, and the inframarginal outlay

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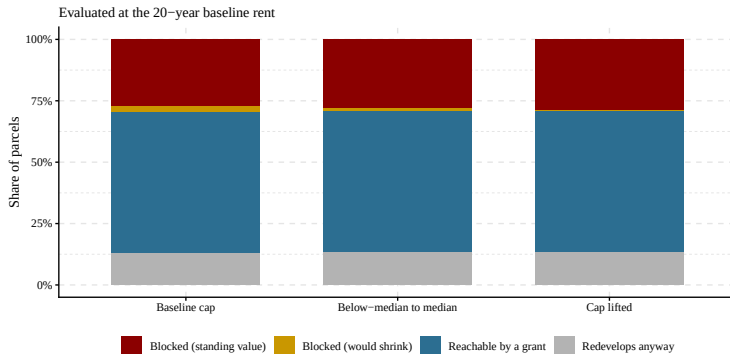
The infeasible budget frontier

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Horizon	Reachable set					1bn EUR budget	
	Reachable	Share	Median $s_{\min}$	$\leq 500/\text{sqm}$	$\leq c_0$	Induced (1bn)	Cost/induced (k)
5	297,469	54.7%	2,309	5.5%	18.7%	7,981	123
10	315,016	57.9%	2,104	8.0%	24.3%	10,697	93
20	310,701	57.1%	1,839	12.7%	32.7%	14,403	68
30	289,833	53.3%	1,714	15.4%	36.1%	15,884	63
50	208,923	38.4%	1,482	19.0%	41.7%	17,826	56

Relaxing the density cap barely moves the stock

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- ▶ Lifting the cap entirely removes two-thirds of cap-blocked parcels, from 12,032 to 3,681
- ▶ For 93 % of those remaining, the unconstrained optimum is *smaller* than the standing structure

Construction time: formal tests

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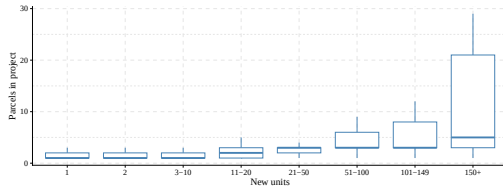
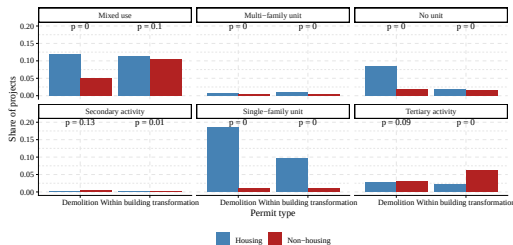
	Annex	Mixed	Multi	New unit	No unit	Secondary	Single
Mixed	0.383						
Multi	0.620	0.058 [*]					
New unit	0.258	<0.001 ^{***}	0.602				
No unit	0.804	0.089 [*]	0.316	0.003 ^{***}			
Secondary	0.038 ^{**}	0.051 [*]	0.010 ^{**}	<0.001 ^{***}	0.020 ^{**}		
Single	0.634	<0.001 ^{***}	0.804	0.089 [*]	0.049 ^{**}	0.003 ^{***}	
Tertiary	<0.001 ^{***}	<0.001 ^{***}	<0.001 ^{***}	<0.001 ^{***}	<0.001 ^{***}	0.708	<0.001 ^{***}

Kruskal–Wallis: $\chi^2 = 118.7$, $df = 7$, $p < 2.2 \times 10^{-16}$

- Kruskal–Wallis rejects equal durations across uses of origin ($\chi^2 = 118.7$, $df = 7$, $p < 2.2 \times 10^{-16}$)

Permits creating housing versus non-housing

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- ▶ 79% of permits involving a demolition supply housing, against 56% of permits without
- ▶ The median project above 150 units requires 5 parcels

Price prediction: hold-out fit and rent validation

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